

Research Plan: Worked Example

The Effect of Background Music Tempo on Typing Speed

About This Document

This worked example demonstrates how to approach the Research Plan assignment using a topic that is NOT one of your assessment options. The topic here—background music tempo and typing speed—follows the same structure you should use for your chosen topic.

Yellow boxes throughout provide commentary on good practice. These annotations explain WHY certain choices are made—pay attention to the reasoning, not just the content.

The content here is abbreviated, in note format (e.g. bullet points) in some places, and is NOT the final polished version I would submit. This is an example of a working document where I draw together all the key information. See the submission example for a better idea of what a good submission would actually look like.

Submission Format

Your submission should include the following elements:

- Title page with: title, Student ID (anonymous), module code, institution, term, word count, and AI declaration
- APA-style section headings (Method, Results, References)
- Method subsections in italics (Design, Participants, Materials, Procedure)
- References page (for your single referenced work)
- Double-spacing and Times New Roman 12pt font throughout



A note on references

We require you to include a single reference to a real piece of research relevant to the chosen topic. This is to assess your ability to 1) create an in-text citation, and 2) to create a correct and complete APA 7 format reference entry. Do NOT write an introduction or background section—just a brief contextualising sentence is sufficient.

A. Introductory sentence & Experimental Hypothesis

Before writing your hypothesis

Think through: What is my IV? What is my DV? What direction do I predict, and why? Your hypothesis should flow naturally from these decisions. If you want to briefly contextualise why your research question matters, one sentence with a citation is enough.

[SUBMISSION STARTS HERE – NO INTRODUCTION HEADING]

Typing speed is an increasingly valuable workplace skill, with research suggesting that efficient keyboard use is associated with reduced occupational frustration and improved job satisfaction (Qwerty, 2025).

The present study investigates whether background music tempo influences typing performance. *It is hypothesised that participants will type significantly faster (more words per minute) when listening to fast-tempo music compared to slow-tempo music.* This is a one-tailed (directional) hypothesis. A directional prediction is appropriate because previous research on arousal and performance suggests that faster tempos increase physiological arousal, which may facilitate faster motor responses in simple tasks like typing.

What makes this hypothesis section effective?

Notice: (1) The opening sentence provides brief context with a citation—this is OPTIONAL but shows you can do it correctly. (2) The actual hypothesis (italicised here for clarity) specifies both the IV and DV. (3) The direction is stated and justified. You do NOT need a full literature review—one contextualising sentence is plenty. The reference will appear on a References page at the end.

B. Method

Design

An independent groups (between-participants) design was used. The independent variable was background music tempo, with two levels: fast tempo (140 beats per minute) and slow tempo (60 beats per minute). The dependent variable was typing speed, operationalised as the number of words correctly typed per minute.

An independent groups design was chosen rather than a repeated measures design for two reasons. First, a repeated measures design would introduce order effects—participants might improve through practice or fatigue over two typing sessions, confounding the effect of music tempo. Second, participants might guess the hypothesis if exposed to both conditions, potentially altering their behaviour (demand characteristics). Random allocation to conditions controls for individual differences in baseline typing ability.

Design justification is crucial

This section demonstrates: (1) Clear statement of design type. (2) IV with SPECIFIC levels (not just 'fast vs slow' but exact BPM values). (3) DV with clear operationalisation (words per minute, not vague 'typing speed'). (4) Genuine justification—explaining WHY this design suits THIS study, not generic advantages.

Participants

Forty participants (20 per condition) will be recruited via opportunity sampling from the undergraduate population at Goldsmiths, University of London. Participants will be aged 18-30 years to reduce variability in typing experience associated with generational differences in computer use.

An approximately equal distribution of male and female participants will be recruited across conditions. Participants will be randomly allocated to conditions using a random number generator.

Participants will be excluded if they have a diagnosed hearing impairment; a motor impairment affecting hand/finger movement; are professional typists or have formal typing training; or do not have English as their first language (as the typing task involves English text).

Participants will be naive to the experimental hypothesis. They will be informed only that the study investigates 'factors affecting typing performance' to minimise demand characteristics.

Participant details matter

Good participant sections include: specific numbers (with justification if possible), clear recruitment method, relevant demographics, thoughtful exclusion criteria (with reasons), and consideration of whether participants know the hypothesis. Note how each exclusion criterion relates to the specific study.

Materials/Stimuli (in abbreviated note style!)

Music stimuli: Two instrumental music tracks will be used, matched on all characteristics except tempo:

- Fast tempo condition: 140 BPM electronic instrumental track
- Slow tempo condition: 60 BPM electronic instrumental track
- Both tracks will be instrumental only (no lyrics), in a major key, at moderate volume (65 dB), and 5 minutes in duration

The tracks will be sourced from royalty-free music libraries and pilot-tested to ensure equivalent perceived pleasantness (to avoid confounding tempo with likeability).

Typing task: Participants will type a standardised 300-word passage displayed on screen. The passage will:

- Contain common English words (no technical jargon or uncommon vocabulary)
- Be emotionally neutral in content
- Be displayed in black 14pt Arial font on a white background

Equipment: Standard desktop computer with QWERTY keyboard; over-ear headphones for music delivery; typing software that records words per minute and errors; stopwatch; consent forms; debrief sheets.

Stimulus control is essential

Notice how EVERYTHING is specified and controlled. The music tracks are matched on key, volume, duration, and pleasantness—only tempo differs. The typing passage is standardised. This level of detail shows you understand that uncontrolled variables become confounds. Ask yourself: 'What else could explain any difference I find?' Then control for it.

Procedure (the real submission must NOT use bullet points)

Participants will be tested individually in a quiet laboratory room. The procedure will follow these steps:

1. Upon arrival, participants will read an information sheet explaining that the study investigates factors affecting typing performance. They will sign a consent form.
2. Participants will be randomly allocated to either the fast-tempo or slow-tempo condition.
3. Participants will be seated at the computer and fitted with headphones. They will complete a brief practice typing task (30 seconds, different passage) to familiarise themselves with the keyboard and software.

4. Participants will be given standardised instructions: 'You will hear some music through the headphones. While the music plays, please type the passage shown on screen as quickly and accurately as you can. The task will last 5 minutes.'
5. The music will begin, and the typing passage will appear on screen. Participants will type for exactly 5 minutes.
6. At the end of 5 minutes, the music will stop and participants will be asked to stop typing. The software will automatically record words typed per minute.
7. Participants will be thanked, fully debriefed about the true purpose of the study (investigating whether music tempo affects typing speed), and given the opportunity to withdraw their data. Contact details for the researcher will be provided for any follow-up questions.



Replicability is the goal

A good procedure section means someone else could run your study and get comparable results. Include: specific timings, exact instructions (in quotes), order of events, what happens at each stage, and ethical elements (consent at start, debrief at end). The practice task is a nice touch—it reduces variability from participants 'warming up' during the actual task.

C. Draft Results Section

Summary Data Table

Table 1 shows the structure for summarising the data. Values are left blank as no data have been collected.

Table 1. Mean typing speed (words per minute) by music tempo condition

Condition	N	Mean (WPM)	SD
Fast tempo (140 BPM)	20		
Slow tempo (60 BPM)	20		

Note. WPM = words per minute. SD = standard deviation. Values left blank as this is a research plan.

Table structure matters

Your table should show: (1) Clear condition labels matching your IV levels. (2) Sample size per group. (3) Appropriate descriptive statistics for your DV (Mean and SD for continuous data). (4) Units specified. (5) A note explaining abbreviations. Do NOT invent numbers—the blank cells show you understand this is a plan. Review APA format tables. Note the above table is NOT APA format.

Statistical Test

The data will be analysed using an independent samples t-test.

Justification: An independent samples t-test is appropriate for the following reasons:

- The design is independent groups (different participants in each condition)
- There are exactly two conditions to compare
- The dependent variable (words per minute) is continuous, ratio-level data
- With 20 participants per group, the central limit theorem suggests the sampling distribution will be approximately normal

Prior to analysis, assumptions of normality (via Shapiro-Wilk test and visual inspection of histograms) and homogeneity of variance (via Levene's test) will be checked. If assumptions are violated, a Mann-Whitney U test will be used as the non-parametric alternative.

Expected reporting format:

'An independent samples t-test revealed a [significant/non-significant] difference in typing speed between the fast tempo condition ($M = \underline{\hspace{1cm}}$, $SD = \underline{\hspace{1cm}}$) and the slow tempo condition ($M = \underline{\hspace{1cm}}$, $SD = \underline{\hspace{1cm}}$), $t(38) = \underline{\hspace{1cm}}$, $p = \underline{\hspace{1cm}}$.'

Match your test to your design

The test choice flows directly from your design decisions: Independent groups → independent samples test. Two groups → t-test (not ANOVA). Continuous DV → parametric test (with non-parametric backup). This section shows you understand the CONNECTION between design and analysis—that's what we're assessing.

D. References (always starts on a new page).

References are optional but must be correct if included

You need only one reference for this assignment. Thus you MUST include a References page with the full citation in APA format. The example below shows correct APA 7th edition formatting: hanging indent, italicised journal title and volume, DOI as a hyperlink (which won't work – it's not a real paper!)

Qwerty, A. (2025). Keyboard proficiency and occupational wellbeing: A meta-analysis. *Journal of Workplace Ergonomics*, 42(3), 215-232.
<https://doi.org/10.1000/jwe.2025.0042>

Summary: What Makes a Good Research Plan?

- A clear, testable hypothesis with justified direction
- Precisely operationalised IV and DV (specific levels, specific units)
- Design choice justified for THIS study, not generic advantages
- Controlled stimuli—everything matched except your IV
- Detailed, replicable procedure with timings and exact instructions
- Statistical test that matches your design and data type
- Thoughtful consideration of confounds and how to control them
- Correct APA formatting if you include any references

Remember: We're assessing your scientific reasoning, not your literature knowledge. Show us you understand WHY you're making each choice.